Page of Contents:

Sl.Order	Particulars
A	Purpose
B	Scope
C	Responsibility
D	Departmental Hierarchy

E	Activity 1. Pharmacy Advisory Committee 2. Essential Drug list 3. **Purchasing** Procedure 4. **Storage** of Medicine in the Pharmacy **Narcotic drug: Usage, administration and disposal** 5. Process of Obtaining the Medicine when the Pharmacy is closed 6. Replenishment of Emergency Medicines 7. **Prescriptions** of Medicines 8. The Essential Drug list- Drug formulary 9. Prescribing High risk Medications 10. **Dispensing** of Medicines 11. Recall of Medicines 12. Labeling of Drugs 13. Administration of Medication 14. Self Administration of Medicine 15. Medicine brought from pharmaceutical store outside the Hospital 16. Educating Patients and Family members about safe medication and food drug interactions 17. Monitoring of Patients 18. Adverse Drug Events 19. Minimizing loss and Pilferage 20. Implantable Prosthesis

A. **Purpose:** To provide guideline instructions for effective management of pharmacy services.

B. **Scope:** It covers all activities under the purview of pharmacy services.

C. **Responsibility Person:** Chief Pharmacist, Pharmacist and Nursing Staff

D. **Departmental**

Hierarchy: Administrator

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Chief Pharmacist

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Senior Pharmacist – In charge of Outpatient Dispensary

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Pharmacist

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Pharmacy Attendants

E. Activity:

1. <u>Pharmacy Advisory committee:</u>

The Pharmacy Advisory committee is a multidisciplinary committee responsible for the formulation and implementation of policies and procedures related to the pharmacy services of the hospital.

The committee monitors all pharmacy related activities and ensures that the pharmacy services of the hospitals functions in compliance with the applicable laws and regulations.

2. <u>Purchasing procedure:</u>

The medical store is responsible for purchase of all pharmaceutical and surgical products required in the hospital for the treatment of patient. All products are purchased based on the reorder level calculated for each product. Pharmacy purchases are done through:

i. Self Purchase

- Few pharmaceutical items are purchased by the hospital directly from the authorized vendors.
- Purchase orders are issued directly to the authorized vendors by the pharmacy department of the hospital depending upon the reorder level for the desired product.
- All payments in this regard will be paid by the hospital from its approved budget

ii. Emergency Purchase:

Emergency purchase is made from identified medical store selected on the basis of the following guideline:

1. Should be located in close proximity of the hospital.

2. Should provide service on an 24 hour basis.

3. Should be ready to offer the product at a cost lower than the maximum retail price (MRP).

4. Should have a good track record in terms of quality of products ,honest practices etc .

Purchasing procedure:

1. Emergency purchase will be made only when there is an immediate need for a product which is not possible to purchase immediately from the usual identified sources.

2. During normal working hours of the hospital the Medical Superintendent has the right to undertake an emergency purchase from the identified pharmacy shop upto a specified amount.

3. In his absence, the administrator is authorized to undertake an emergency purchase upto the specified amount.

4. After the normal working hours when the CMS and the MS is not available , the on duty Emergency Medical Officer is authorized to make an emergency purchase upto an amount of Rs 500.

5. No other hospital employee is allowed to make emergency purchase without the written permission from the above mentioned authorities.

4. Storage of Medicines in the Pharmacy:

All pharmaceutical items are arranged alphabetically in the racks. Tablets, Injections, ointments syrups, inhalers, surgicals etc are stored separately in racks so that easy access is facilitated

The items which are to stored at 2*c to 8*c are arranged in the refrigerator provided which is connected to UPS line to maintain the cool chain .Regular temperature atleast twice a day is documented in the designated register (Ref Register #)

There is separate cupboard for keeping emergency drugs and high risk medication.

A room thermometer is fixed in the pharmacy to check the temperature

Narcotics are not stocked in the hospital. A record of use, administration and disposal of narcotic drugs is kept by pharmacy incharge and ICU incharge separately.

A list of **sound alike and look alike** medicines are updated by the pharmacy incharge and the same are **stored separately**. Staff handling such drugs is informed about their usage.

Narcotic drug: Usage, administration and disposal

Narcotics are not stocked in the hospital. A record of use, administration and disposal of narcotic drugs is kept by pharmacy incharge and ICU incharge separately.

Unused narcotic tabs are returned to the pharmacy from where it was procured and they will dispose. Drugs such as powerful narcotic pain relievers and other controlled substances carry instructions for flushing to reduce the danger of unintentional use or overdose and illegal abuse.(Toilet for tablets, sink for liquids.)

For example, the fentanyl patch, an adhesive patch that delivers a potent pain medicine through the skin, comes with instructions to flush used or leftover patches. Too much fentanyl can cause severe breathing problems and lead to death in babies, children, pets, and even adults, especially those who have not been prescribed the drug. “Even after a patch is used, a lot of the drug remains in the patch, so you wouldn’t want to throw something in the trash that contains a powerful and potentially dangerous narcotic that could harm others.”

Controlled substances (e.g. narcotics and psychotropics) require tight security and control. In

some countries, scavenging of material from landfills is a frequent problem, and, disposed drugs may be recovered and sold by the scavengers. Measures are therefore necessary to prevent diversion during sorting, and pilfering of drugs from landfills. Immobilization is the best method of preventing pilfering from a store or landfill. If as a last resort, pharmaceuticals must be discarded direct to a landfill then they must be covered immediately with a large quantity of municipal waste.

5. <u>Process for Obtaining Medicine when the pharmacy is closed:</u>

The hospital pharmacy operates on a 24 basis. Medicines required for each admitted patients are stored on a daily basis in the respective wards and in the emergency department. In case of urgent need of any drug that is not available in the hospital, the on duty Emergency Medical Officer will purchase the product from identified pharmacy shop.

6. <u>Replenishment of Emergency Medicines:</u>

Reordering level for all emergency drugs is made separately and that is checked every day by the senior pharmacist if any thing less then the reordering level, it will be replenished immediately by placing orders with appropriate authorities.

List of emergency drugs is made by pharmacy incharge and that list is kept in the pharmacy for reference.

7. Prescription of Medicines:

Medicines can be prescribed only by doctor associated with the hospital as a visiting consultant or resident doctor.

Prescription given by the outside medical doctor will also be honoured in the hospital.

All medication orders are to be prescribed in writing which should be dated, timed, signed by the prescribing doctor. The Essential Drug list- Drug formulary is followed while prescribing medicines.

It is the policy of the hospital that all prescriptions comply with the law, and contain details that can be clearly interpreted.

No drug will be administered to a patient without a valid prescription of treating doctor in an emergency when a consultant is contacted on phone and the drug is prescribed by him, the Medicine may be given to the patient under the signature of the locally available treating doctor and this should be authenticated by the Prescribing Consultant within 24 hrs.

All prescriptions should have the following details

a) Patient's name, registration number (OP and IP as applicable).
b) Drug name (generic names or trade names written in full), strength, dose and form.
c) Frequency of administration of medicines, indicated by clear and definitely stated intervals.
d) Duration of treatment i.e. upto validity period of OPD ticket (15days to 30 days).
e) Doctor's signature and date.
f) No medications may be dispensed for prescriptions that do not have the above mentioned details.
g) Drug names should be generic names or trade names written in full. If abbreviations are used, the doctor must be contacted for confirmation.
h) Certain prescribing conventions are desirable, e.g. underlining and initialing an unusual quantity, strength or directions.
i) In case the contents of the prescription are not clear, clarification should be obtained from the concerned doctor before dispensing.
j) Alterations and cancellations must be initialed by the doctor.
k) Only approved conventions and abbreviations should be used.
l) Metric: Medication orders shall be written in metric notation only. e.g., mg, gm, ml.

m) PRN: Orders for "as needed" or "PRN" medications shall specify the dose, dosage form, and duration and dosage frequency.
n) Renewal: The use of the terms "renew", "repeat" in reference to previous orders are not acceptable ,can write continue same treatment (CST).
o) Therapeutic Substitution: In limited, low risk, high volume cases certain over-the-counter groups of drugs or products may be substituted for different drugs or products.

Examples of such items are enteral formulae, liquid antacids and multivitamins.
Prescriptions from sources outside the hospital shall be dispensed by the pharmacy

8. The Essential Drug list- Drug formulary :

The hospital has a list of medications appropriate for the patient's and organisation's resources.

9. Prescribing High Risk Medications:

The hospital has identified a list of High Risk Medicines such as :

I. Intravenous Potassium
II. Methotrexate
III. Hypotonic /Hypertonic sodium chloride
IV. Inj adrenaline
V. Inj aminophyllin
VI. Inj atropine
VII. Inj atracurium
VIII. Inj botophase
IX. Inj calcium gluconate
X. Inj cardrone
XI. Inj diazepam
XII. Inj digoxin
XIII. Inj dopamine
XIV. Inj dobutamine
XV. Inj ephedrine
XVI. Inj epsolin

XVII. Inj fentanyl
XVIII. Inj heparin (low molecular weight)
XIX. Inj insulin
XX. Inj magnesium sulphate
XXI. Inj midazolam

XXII. Inj nitroglycerine
XXIII. Inj noradrenaline
XXIV. Inj pethedine
XXV. Inj propofol
XXVI. Inj serenace
XXVII. Inj streptokinase
XXVIII. Inj sodium valporate
XXXIX. Inj thiopentone
XXXX Inj vecuronium

The List is evaluated by the Drug and Therapeutic committee at periodic intervals. High Risk Medicines are to be prescribed in writing by the concerned doctor. Prior to dispensing of high risk medicines the written order is verified by the pharmacy incharge.The pharmacy incharge/pharmacists verifies the following prior to dispensing of High Risk Medicines:

1. Name of the Medicine
2. Quantity and Dose Prescribed
3. Name of the Prescribing Doctor with his signature, date and time.

Only after confirming the same the drugs is issued from the pharmacy .

10. <u>Dispensing of Medicines:</u>

Dispensing of medication is done in a manner that ensures quick and efficient patient care and minimizes errors.

A. Inpatient Dispensing:

1) Pharmacy items are dispensed / issued only by a Pharmacist to nurse in charge of unit. Drug administration to patient is done by nursing staff in wards. No self- medication is allowed in ward.
2) Stat doses and discharge medications will be given first priority for dispensing.
3) Prescriptions/Indent must be read carefully; the signature of the prescribing doctor must be verified.

4) Correct Dose and Dose form for each individual patients are checked by another staff nurse prior to dispensing of medicines.
5) Items are collected from the designated racks, storage etc as applicable.
6) Any item prior to dispensing are checked for the expiry date.

7) Post Dispensing the transaction is entered in the issue register.
8) Entries relating to dispensing of Narcotic Drugs are entered in the specified column of the Narcotic Drug Register along with the name of the drug and prescribing doctor name of the patient along with the UHID ,date ,quantity issued etc

B. Outpatient Dispensing:

Dispensing of pharmaceutical items to outpatient are done from the hospital Dispensary.

1) Prior to dispensing the Medicine the pharmacist verifies the following details in the prescription :

a) Name of the Patient, registration number
b) Name of the drugs, dose and route prescribed.
c) Name and Signature of the Prescribing Doctor along with the date and time.

2) Expiry Date of the Item is checked prior to dispensing of the item by the pharmacist.

3) Items from the dispensary are dispensed only by the concerned pharmacist.

4) Prior to Dispensing the dosage for individual drugs are explained to the patient , special precautions if any like for example :

a) To be taken before food
b) To take plenty of water
c) To complete the full course for antibiotic are clearly explained to the patient/relatives by the concerned pharmacist.

11. <u>Recall of Medicines</u>:

1. On receiving any complaints from the wards on medicines and surgicals items like medicines, Examples

A) Problem of dissolution of the dry powder when reconstituting the injection
B) Problem of discoloration or different colour after reconstituting the vial
C) A suspended impurities noted on reconstitution
D) A suspended particle in LV.fluids etc

In any of the above event the ward nurse or head nurse will immediately report to pharmacy:

1) The same is confirmed by the pharmacy incharge

2) Immediately a letter is issued to the concern wards and departments where the specified medicine has been issued

3) The same is communicated immediately to the concerned supplier immediately and a feed back of the same is requested after investigation.

12. <u>Labeling of Drugs:</u>

It is the policy of this hospital that all drugs and medications maintained in the hospital is properly labeled.

Drugs labels must be legible, clear and consistent at all times.

Any drug label soiled, incomplete, illegible, worn, or make shift must be returned and replaced by the pharmacy.

The following details must be on all cut tablets/capsules strip or loose tablets dispensed for inpatient:

1) Name of the medicine,
2) Strength and form of medicine
3) Quantity of medicine
4) Frequency of Administration etc.

13. <u>Administration of Medication</u>:

The hospital's policy is to ensure proper administration of medicine. Administration of medicine can be done only by Registered Medical Practioners .In addition to the medical practioners, registered nursing staff is also allowed to administer medicines. The following details are verified by the concerned hospital staff prior to administration of drug:

1. Identification of patient is done by confirming the patient's name, unique hospital identification number of the patient in the patient's case record.

2. The treatment orders of the medical practioners are verified to confirm.

a. The name of the medicine (by matching with the treatment order) prior to administration of the same.

b. The specified dose for the medicine

c. Route for administration of medicine example intravenous, oral etc.

d. Time for administration of medicine as indicated in the doctors treatment orders Post Administration of medicine, record of the same are entered in the designated register (maintained in the respective wards) and the patients case record. Record indicates the following:

1. Signature of the staff responsible for administration of the drug.
2. Time at which the drug was administered
3. Dosage and route for administration of the medicine.

The primary treating doctor or the on duty medical officers are responsible to counter check the record to verify appropriateness of administration of medicine. The same is done by interviewing the concerned patient or his/her relatives.

DOTS Therapy for T.B. Patients. under RNTCP (Revised National .T.B. Control Program)

Medication is labeled prior to preparation of a second drug.

14. Self Administration of Medicine:

It is the hospitals policy not to allow self administration of medicine by the patients however in case of acute long standing diseases such as Diabetes where the patient is on self administered medicine etc this should be brought to the notice of the treating consultant and on his / her reassessment patient's treatment will be modified or carried out by the health care providers.

If a patient is on long term drug therapy (oral) his / her continuing the drug in the hospital, will be decided by the concerned consultant. Incases where self administration of drug is allowed, the same is to be indicated in the patients medical record by the primary treating consultant clearly stating the reasons for allowing self administration of drug.

15. Medicine brought from pharmaceutical stores out side the hospital:

All the doctors are required to prescribe medicines following the Essential Drug list- Drug formulary. However in cases (e.g. Non availability of the drug due to shortage of supply, incase of emergency etc) where the medicine is to be purchased from outside, it is mandatory for the hospital staff to check the label of the medicine to check the name of the drug, its expiry date etc. prior to administration of the drug.

16. Educating patients and family members about Safe Medication and Food drug interaction:

The hospital identifies the importance of educating patients and family members about safe medication and food drug interaction for facilitating quick recovery of the patient.

The patient/relatives are clearly explained about the required dosage, the time interval at which the medicine is to be taken , special precautions in terms of food like whether the medicine is to be taken on an empty stomach or any diet restrictions , special diet schedule to followed etc by the treating doctor at the time of prescribing the medicine. The same is indicated in the prescription/patients case record in clear legible writing by the treating doctor.

In the hospital dispensary at the time of dispensing medicine the concerned pharmacist re-educates patients /relatives about safe medication practices in relation to the prescribed medicines. Incase any special precautions to be taken in terms of food the same is explained to the patient/relatives.

Patients are encouraged to get the drug verified from the concerned consultant doctor or the intern doctor after collecting it from the dispensary.

For inpatients, the ward nurse must verify the medicine properly prior to its administration.

In the respective wards the concerned nursing staff also educates the patient/relatives about safe medication practices including dosage of the prescribed drug, time interval at which medicines is to be taken, special precautions in terms of diet if any to be followed etc.

17. Monitoring of Patient:

The hospital strives towards speedy recovery of all patients hence it recognizes the importance of proper administration of medicine in facilitating quick and effective recovery of the patient. Post administration of drug the monitoring of patient is done by the concerned medical and nursing staff to mark the progress of patient at periodic intervals interms of the drug administered. Monitoring is also done to evaluate any adverse event in terms of the drug administered.

18. Adverse Drug Events ;

An Adverse Drug event can be defined as any unfavorable and unintended signs including an abnormal laboratory findings symptoms or disease temporarily associated with the use of drugs

Adverse reaction may be :

1. Dermatological reaction like mild skin rashes

2. Respiratory like brocho spasm, respiratory depression

3. Blood, blood dacryasis

4. Congenital Anomaly

5. Hepatic liver enzyme elevation, liver cirrhosis

6. Death.

Adverse Drug Reactions, if any are reported immediately to the patient's attending Doctor, Nursing matron and Nursing incharge, pharmacy incharge. The incident is immediately recorded in the Adverse Drug Events Reporting form (ref #) available in every patient care area of the hospital.

The reported formats are forwarded to the Drug and Therapeutic Committee who meet at periodic intervals to evaluate such events in order to analyze such adverse drug events to monitor the risk, rate and trend of such reactions and suggest measures to prevent such events in future. Incase unacceptable trends are reported the Drug and Therapeutic Committee in consultation with the Chief Medical Superintendent can modify the policy as per the need of the situation to reduce such Adverse Drug events.

19. Minimizing cost and Pilferage: See General Store

20. Implantable Prosthesis:

The hospital has layed down guidelines for safe procurement and usage of Implantable prosthesis as indicated below:

i. Procurement and Usage of Implantable prosthesis:

1. **Aim:**
   To provide uniform guidelines for purchase of implants.

2. **Rationale:**
   - Purchase of the most cost-effective implants in the right quantities.
   - Selection of reliable suppliers of high-quality products.
   - Ensure timely delivery.
   - Achieve the lowest possible total cost.

3. **Personnel responsible :**
    - Director
    - Medical superintendent
    - OT Incharge

4. **Procedure**

- implants selection, which is done by a formulary or essential implants list.
- Quantification of implant requirements, which should have inputs from the OT Incharge.
- Product specifications, which should be prepared by Medical superintendent or Director.
- Selection of suppliers, which should be done by director, medical superintendent.
- Quotations are take for selection.
- Order- It is done by the approval of the director.

Implants Requisition is prepared by OT nursing in charge as needed by eye surgeon for intraocular lens. It is signed by Head – Department of Ophthalmology. Implants are issued from the Medical Store as and when indented by the Operation Theatre nurse.

Implant related to Orthopaedics may be purchased by the patient party upon instruction by the consultant Orthopaedic surgeon.

However prior to the use of the implant purchased by the patient party, the same is thoroughly examined by the Orthopaedic surgeon to determine its adequacy and quality for the patient.